London Business School

Mergers, MBOs & Other Corporate Reorganizations

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Paramount
TM
N
WB
TM

- Paramount extended the deadline on its tender offer for Warner Bros. Discovery to Feb. 20 and urged shareholders on Thursday to reject the latest offer from rival suitor Netflix.
- The extension gives Paramount more time to court Warner shareholders with its $77.9 billion offer for the full company.
- Paramount stock rose 1.6% on Thursday
- WB filed SCHEDULE 14D-9 from with SEC last week

Valuation Takeaways

Feature	**Netflix Offer**	**Paramount Offer**
Cash per Share to WBD	$27.75 (all cash)	$30.00
Enterprise Value	~$82.7 B	~$108.4 B
Additional Value	Shares in Discovery Global spin-off	None specified beyond cash
Financing Certainty	Fully committed, less leveraged	Heavily debt-financed, contingent equity
Regulatory Complexity	Targeted assets deal (studios/streaming)	Whole company deal — higher scrutiny

1. Netflix Offer — Structured Terms

- Merger consideration: $27.75 per WBD share, paid entirely in cash under the amended Netflix Merger Agreement.
- Discovery Global spin-off: WBD shareholders will also receive shares in the Discovery Global entity (linear networks like CNN, TNT Sports, Discovery+), creating additional value beyond the $27.75 cash.
- Financial analyses cited estimated implied equity values for Discovery Global in various ranges for spin-off valuation.
- The Netflix deal reduced net indebtedness allocated to Discovery Global by ~$260 million, enhancing value certainty.
- $5.8 billion reverse termination fee protection if regulatory approval fails

2. Paramount Offer — Higher But Riskier?

- Paramount's hostile bid offered $30 per share in cash, valuing the entire company at roughly $108 billion on an all-cash basis.
- By contrast, Netflix's deal valued the acquired assets at ~$82.7 billion enterprise value (~$72 billion equity value).
- Paramount's own complaint (in separate SEC filings) argued that its $30 cash bid was "superior" simply on per-share cash value versus Netflix's $27.75 cash-plus-stock (before Netflix amended to all-cash) plus spin-off value.

- **Why Warner Bros. Discovery's Board Still (Unanimously) Preferred Netflix**
- **1. Certainty and Structure of Value**
- The Netflix transaction, **especially after converting to all-cash**, provided *more certain value* compared to Paramount's offer.
- Paramount's bid, while higher per share, **relied heavily on debt financing** The Netflix structure, including the spin-off, gave shareholders exposure to potential future value in Discovery Global.
- **2. Comparative Risks of Paramount's Financing**
- The board cited **significant risks and uncertainties** in the Paramount proposal:
- Paramount's financing depended on a *complex mix of debt and equity-backstop commitments* (including from third parties such as the Ellison family and others), raising concerns about *certainty of closing*.
- **3. Regulatory and Strategic Considerations**
- Paramount's full-company bid (including linear networks) could face more intense regulatory scrutiny and risk to closing compared with Netflix's offer for studios and streaming.
- The Netflix deal allowed shareholders to retain ownership in the cable/network assets, giving *strategic and financial flexibility*.

Boomberg News:

Paramount has done an effective job of muddying the waters, creating a lot of confusion about what is actually happening. It's also done a good job of annoying Warner Bros. and Netflix.

But most of the recent maneuvers are just noise. The one tangible change was that Paramount prompted Netflix to alter its bid to all cash. But ***unless Warner Bros. shareholders reject the Netflix deal — a long shot — Netflix has won.***

That doesn't mean Paramount is out of options. ***The most obvious step would be to offer more money.*** *The Ellisons have lots of money and Netflix's shareholders are already uncertain about this transaction. Paramount has said its offer isn't its last and final, but it has yet to increase its bid.*

Even though Paramount believes its current bid to be superior to the Netflix terms, the Warner Bros. board doesn't agree. Paramount can offer more money at any point before Warner Bros. shareholders vote on the Netflix deal. After they vote, it gets more complicated.

Paramount could also wait and hope regulators block the deal*. Many politicians have raised concerns about Netflix buying the Warner Bros. studios and HBO, which at a minimum will receive an extensive review in the US and Europe. This is a riskier approach — if the governments approve the deal, there's little Paramount can do. Meanwhile, Paramount's cable networks will shrink even more over that time.*

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Session 4:

Bidding tactics, hostile takeovers, and regulation

Overview of class

- Bidding tactics
- Hostile takeovers

- Regulatory aspects
  - Securities law
  - Antitrust law
  - Foreign investment laws

Merger Negotiations: First contact

- Buyer starts by contacting the target management
  - Often through investment bankers

- Confidentiality is key: Rumors increase the target's stock price
  - The parties cannot deny the existence of merger negotiations if they exist (e.g., U.S. Supreme Court decision)
  - Bankers, lawyers sign confidentiality agreements to start due diligence and access documentation

- Buyer wants to reach agreement as quickly as possible...
- … seller wants to buy time
  - to extract a better price
  - To get other bidders interested

Merger Negotiations: Target reaction

- Target's board forms special committee to evaluate proposal
  - Directors who might profit directly should not take part
  - Hires investment bank to help evaluate transaction terms and price

- Remember: board of directors must make decisions in best interest of shareholders, and no one else
  - In particular: not the executives, not the employees, etc.
  - "Fiduciary duty"
  - Failure to do so exposes the board to lawsuits

- If friendly agreement is reached: press release, information disseminated to the market…
- … but this is only one possibility of course!

Tender offers

Tender offers can be made:

- As part of a final negotiated deal
  - Target board of directors recommends that shareholders tender
  - The deal is “friendly”
- At the outset or in the middle of an existing negotiation
  - To put pressure on the target board of directors
  - Tactic known as a “bear hug”
  - Often, a price is disclosed publicly
- Without any formal warning or contact to the target
  - If the target’s board rejects the offer: hostile deal

The “disappearance” of hostile deals in the US

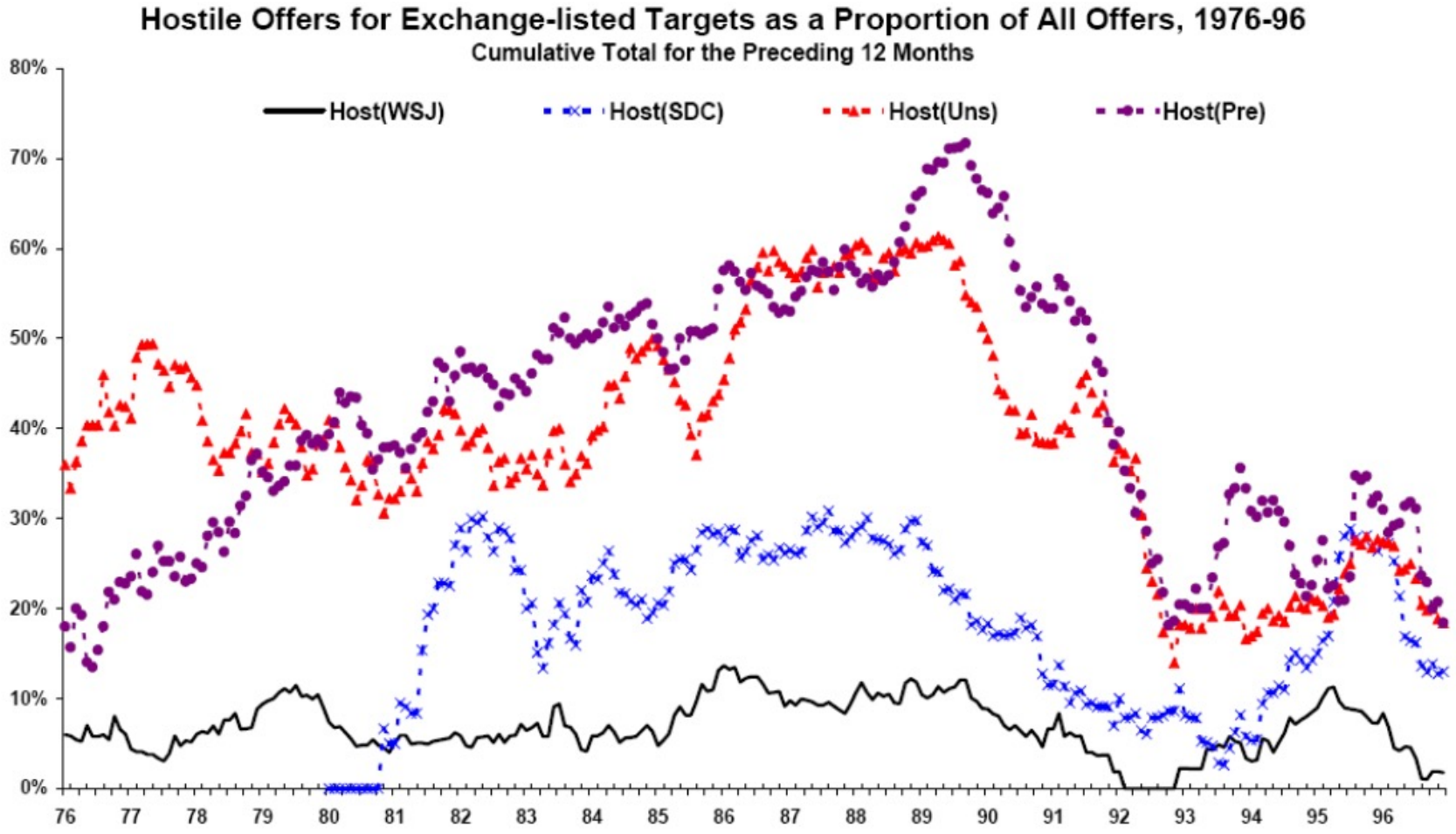


Figure 1. Proportion of all merger or tender offers that are perceived as hostile, by year and by four measures of hostility, for exchange-listed target firms in the period 1975-96. Hostile offers are measured as all those ever explicitly characterized as such in a Wall Street Journal or Dow Jones News Retrieval story [Host(WSJ)], all those characterized as such by Securities Data Corporation [Host(SDC)], all those where the initial or the winning offer is unsolicited [Host(Uns)], and all those where during the prior year there was an initial Form 13D filing or a WSJ or DJNR story about takeover rumors [Host(Pre)].

Source: Schwert (2000,JF)

The Disappearance of Hostile Deals (US)

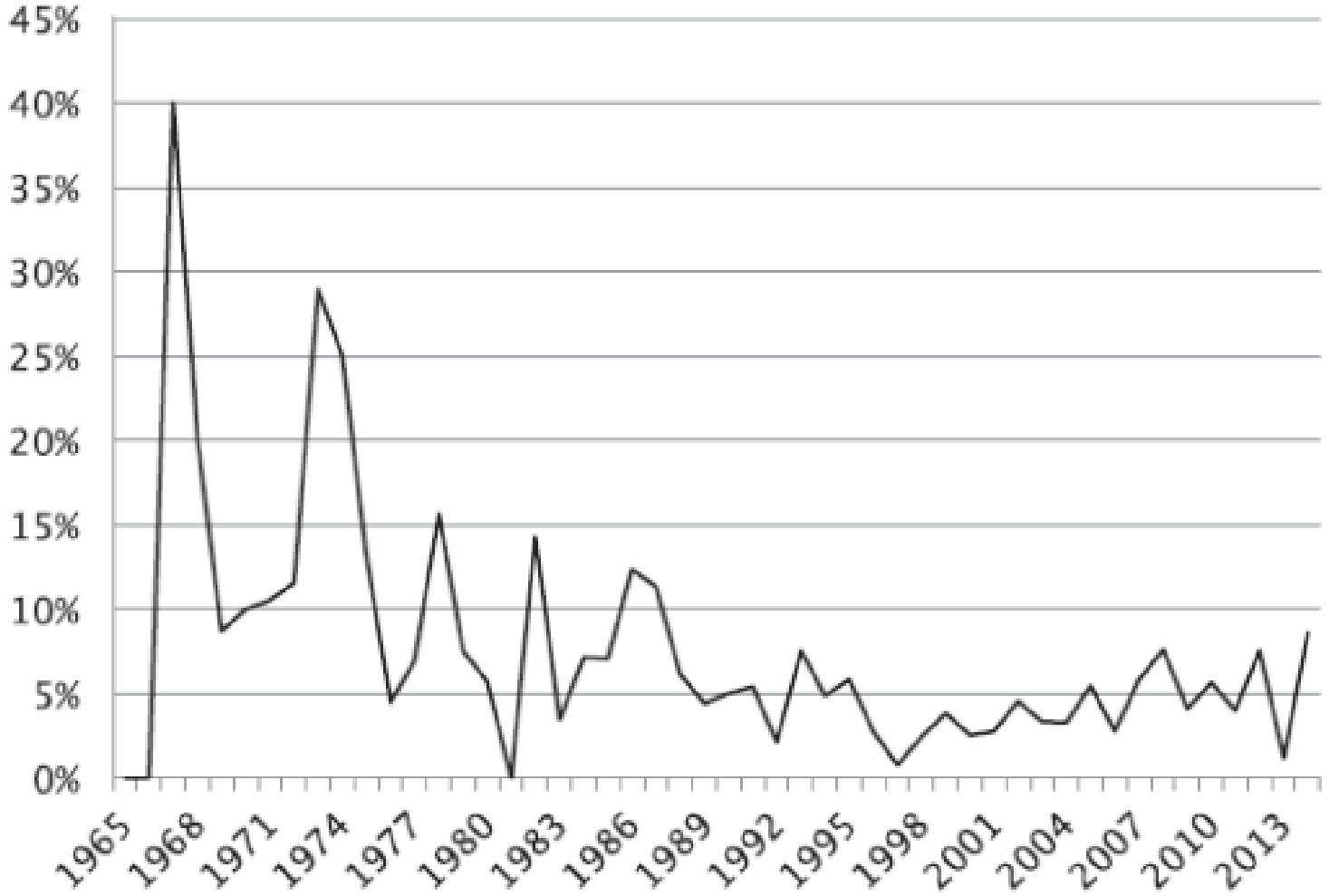


Fig. 2. Conditional on being acquired in a given year, probability of hostility (equal-weighted). Firm-years are coded zero for friendly takeover and one for hostile takeover. Firm-years with no takeovers are excluded.

Source: Cain, McKeon, Solomon (2017,JFE)

Why few(er) hostile deals?

- Hostile deals often not a good deal for bidder
  - Schwert (2000)
  - Higher average premiums
  - Lower success rates
  - Higher likelihood to involve multiple bidders

- Targets gradually came up with better defenses
  - Encouraged by changes in regulation discouraging hostile deals

What takeover defenses?

Ordinary business:	*Pre-bid*	*Post-bid*
Pure defence	Poison pills	Poison pills
	Dual share classes	Pac-man defence
	Voting ceiling	
	Corporate charter amendments	
	Staggered boards	
Embedded defence	Change of control provisions in joint venture, debt, customer and employment contracts	Spin-offs (crown jewels)
		White knight/squire
	Strategic acquisitions	Anti-trust litigation
	Golden parachutes	Strategic acquisitions
	Tin parachutes	Share buy-backs
		Special dividends

Source: Adapted from Arlen and Talley (2003)

Defenses provided by the law (US)

Williams Act	The 1968 amendment to the Securities and Exchange Commission Act of 1934 to regulate tender offers. Requires Securities and Exchange Commission filings, disclosure, and waiting periods. Applies equally to all firms in all states.
1st Generation Statutes	Requires state filing and review requirements. Allows target firm executives, directors, or state commissioners to seek a hearing to delay or block a proposed takeover. The US Supreme Court struck down the Illinois statute (and all other states' statutes by implication) in its *Edgar v. Mite* decision in 1982. Several states struck down their statutes before or after this (*1st Gen Case* in Table 2).
Business Combination	Also known as freeze-out statutes, prohibits bidders from engaging in a business combination with a target for a pre-set period upon the bidder's acquisition of 20% or more of the target's equity unless the purchase is pre-approved by the target's board or a specified percentage of disinterested target stockholders.
Fair Price	Requires disinterested board approval, a supermajority vote of shareholders (usually 80%), or the payment of a minimum price in any second step merger.
Control Share Acquisition	Any target shares acquired by bidder in excess of a threshold cannot be voted by the bidder unless approved by a majority or supermajority of disinterested target shareholders.
Control Share Cash-Out	Dissident target shareholders gain the right to cash-out or sell their shares to the bidder at the highest acquiring price paid during the acquisition period.

Source: Cain, McKeon, Solomon (2017,JFE)

Defenses provided by corporate law (US)

Poison Pill	If executed, dilutes a hostile bidder's toehold stake significantly. A stronger version, known as a dead-hand pill, can be redeemed prior to expiration only by the directors who adopted the pill even if those directors are removed through a proxy contest.
Mandatory Staggered Board	Requires all firms to have a staggered, or classified, board of directors. Generally one-third of directors stand for reelection each year.
Expanded Constituency	Allow boards to consider welfare interests other than shareholders in their deliberations, including workers, creditors, localities, and social considerations.
Disgorgement	Allows a target to recover any potential profits obtained by a person or group who held more than 20% of the issuer in an 18-month period prior to the takeover.
Anti-Greenmail	Prohibits targets from repurchasing toehold shares from a hostile bidder at a premium to the current stock price.
Golden Parachute Restriction	Limits the ability of a target to issue large severance payments to executives in the event of a successful takeover.
Tin Parachute Blessing	Also known as silver parachutes, allows the target to issue large severance payments to a significant number of employees in the event of a successful takeover.
Assumption of Labor Contracts	Requires a successful hostile bidder to assume all preexisting labor contracts of the target firm after a change in control.

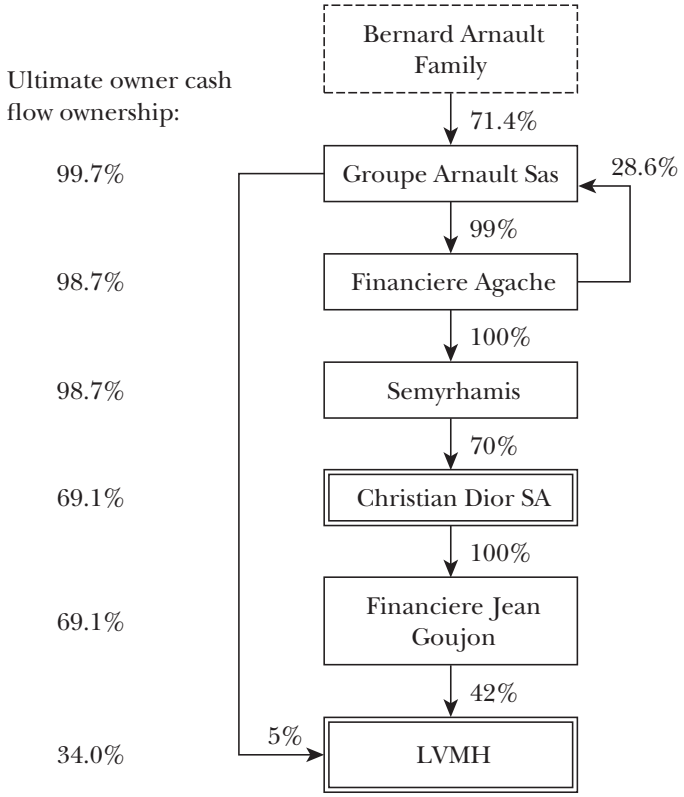
Source: Cain, McKeon, Solomon (2017,JFE)

Other popular takeover defenses in Europe

- Pyramids
- Interlocking structures or boards
- Licensing arrangements
- Capture of foreign investment control authorities

Separation of ownership and control: LVMH

Figure 1
Ownership Structure of LVMH in 2005



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History of Takeover Regulations

Period	*Type*	*Examples*	*Regulatory response*
1897-1904	Creation of large monopolies via horizontal mergers	U.S. Steel DuPont Morgan Bank	Sherman Act (1890) Clayton Act (1914)
1916-1929	Vertical mergers Creation of oligopolies	IBM General Motors Union Carbide	Securities Exchange Act (1934)
1965-1969	Creation of diversified conglomerates Tender-offers	GE ITT Gulf & Western	Williams Act (1968) HSR Act (1976)
1981-1989	Hostile bids Corporate raiders Use of debt to finance very large deals	RJR Nabisco Time-Warner SK Beecham	DOJ merger guidelines (1992) Insider Trading Act (1988) Delaware Takeover Law (1988)
1992-2000	Strategic mergers brought about by deregulation, global competition, technological change	Time / AOL W. Lambert/Pfizer GTE/Qwest	Sarbanes-Oxley (2002)
2003-2008	Cross-Border deals Horizontal mega-mergers LBO jumbo deals Influence of Private Equity buyers	Mittal / Arcelor P&G / Gilette Blackstone / Equity Office Properties	Crackdown on foreign buyers/PE funds in Europe Dodd-Frank (2010)
2014-?	Emerging market buyers Inversions Ultra-low borrowing costs	AB InBev / SabMiller Holcim / Lafarge Actavis / Allergan	Further crackdown on foreign buyers/Trump/Brexit

How Do Firms React to Regulation?

- They may simply abide by it
  - At the expense of failing to complete an NPV > 0 acquisition
  - At the expense of taking remedies such as selling assets or behaving according to law
- They may bypass the regulation to complete an acquisition
- They may creatively use it to prevent an acquisition
- They may lobby to reinforce or decrease the weight of regulation
- All of this is value-relevant:
  - NPV loss from unrealized projects + remedies
  - Lawyer, lobbying expenses, judicial penalties

A taxonomy of regulation

- Securities law
  - Goal: making sure the acquisition is not made at the expense of uninformed and/or small investors in the stock market
  - Regulator: SEC in the US, Financial Conduct Authority (FCA) in UK

- Antitrust law
  - Goal: making sure the acquisition does not lead to a monopolistic position for the acquirer in its market
  - Regulator: DOJ, FTC in the US; Competition and Markets Authority (CMA) in UK

- Foreign investment laws
  - Goal: making sure the acquisition does not lead to loss of control of local country over some “strategic” assets (considered as public goods)
  - Committee on Foreign Investment in the United States (CFIUS) reviews foreign acquisitions for national security risks in US

Securities Law: Limits to Toeholds

- Toehold: bidder accumulates shares in target before official bid
  - Lowers the cost of the acquisition, increases likelihood of success.

- Limit: securities laws may require
  - Disclosure of the purchase of significant positions in a firm's stock
  - Mandatory bids to be made above a certain threshold
  - Example of takeover defense enforced by law

Limits to Toeholds: across Europe

Mandatory bid thresholds	%
France	33.3
Italy	30
Germany	30
Spain	25
United Kingdom	30

Source: Grant and Kirchmaier (2004)

Antitrust regulation

- Competition laws. Antitrust body must review the merger

- Example for the EU:
  - Filing if worldwide turnover > €5Bn and EU turnover > €250M
  - Change in control, where control means “significant influence”
  - 25 working days for a Phase I, 90 days if Phase II (in-depth)

Antitrust regulation: what's the theory?

- Horizontal mergers have a detrimental effect on prices/quality…
  - … unless efficiency gains are large enough
  - … and the higher the merging parties' market power the larger the cost savings needed not to have anticompetitive effects

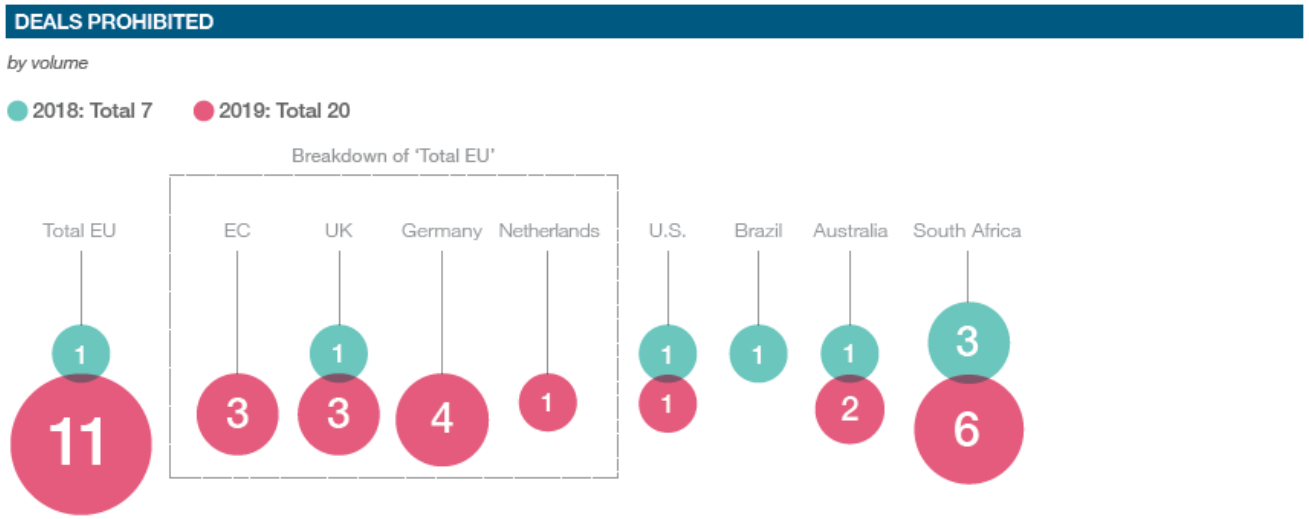
- Vertical and conglomerate mergers: so far considered less likely to harm competition

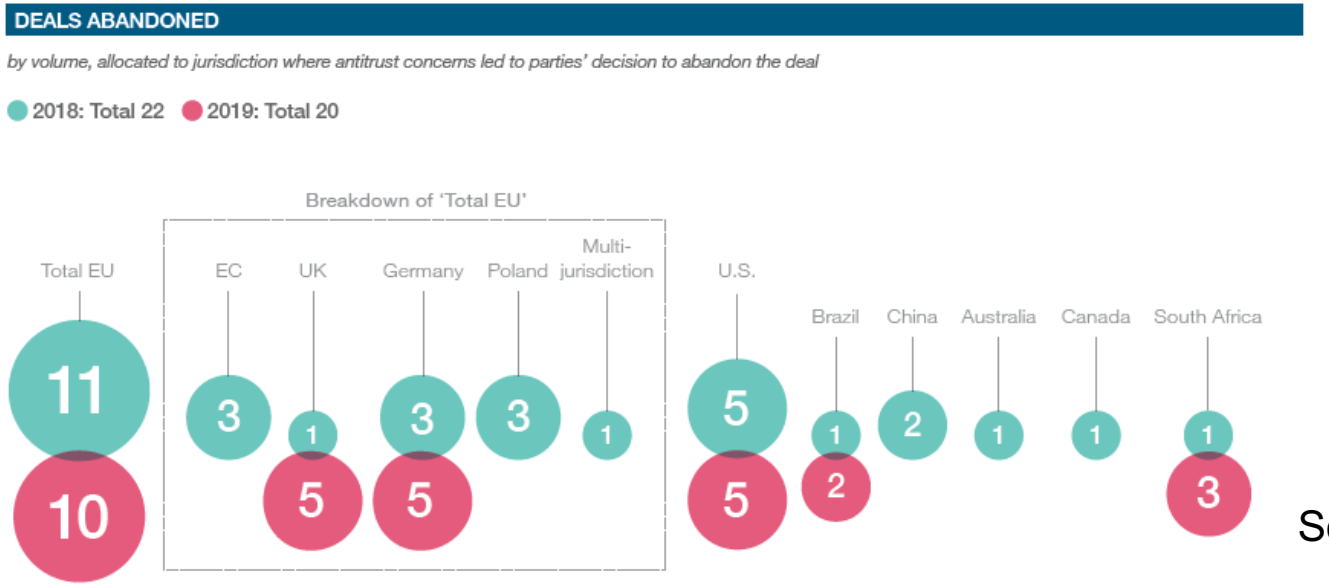
- It is competition agencies which must show a merger "substantially lessens competition"

Antitrust regulation: is it needed?

- Sectoral concentration has increased
- Mark-ups have been rising considerably in the last decades
- Firms' profitability has increased, and its distribution is more unequal
- Common ownership may reduce rivals' incentives to compete
  - Common ownership = a few funds (think Blackrock) own very large part of stock market, including in two firms competing with each other
- Weak merger (and antitrust) enforcement may be one of the factors contributing to this trend (which signals lower competition)

Antitrust regulation: frequency



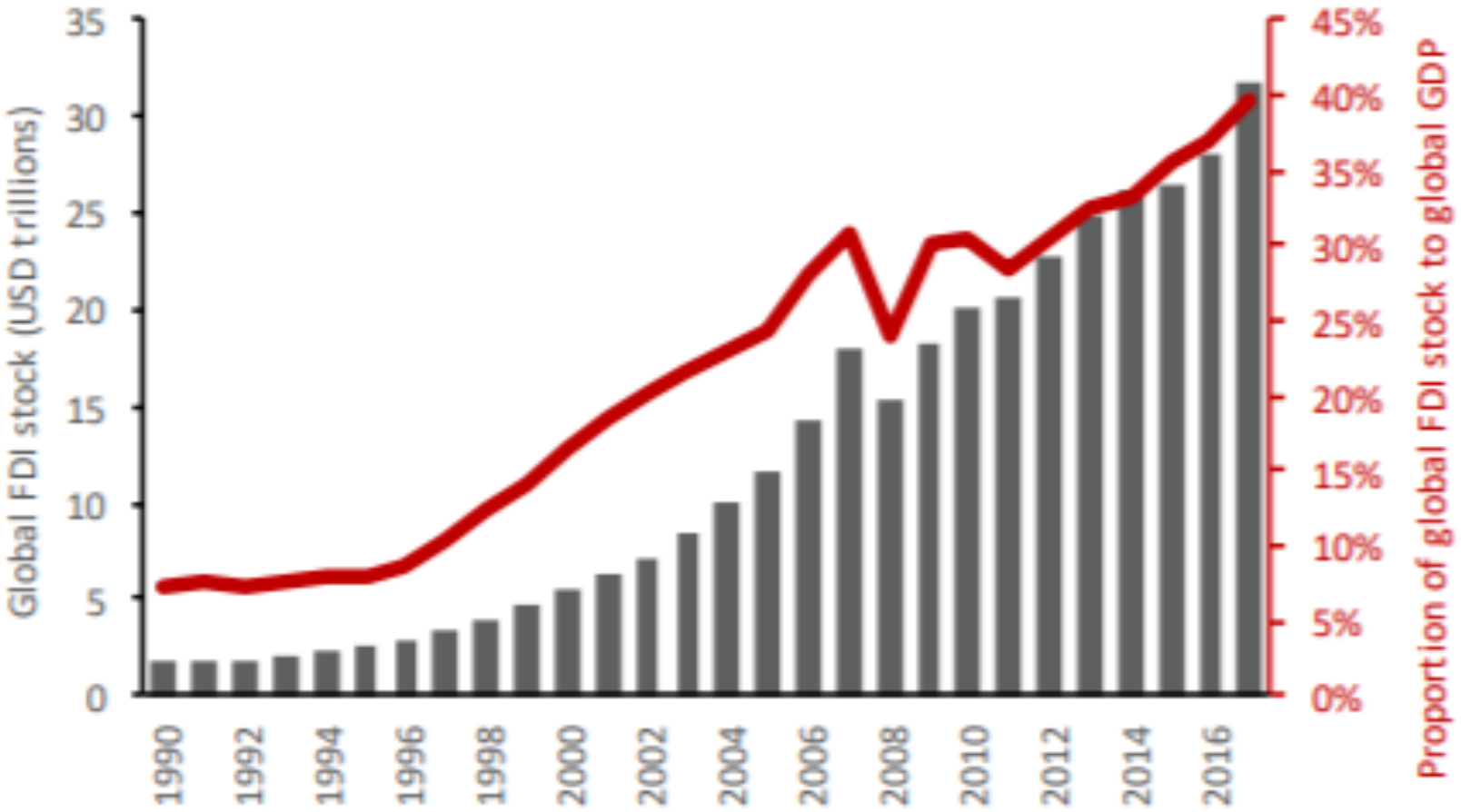


Source: Allen and Overy (2020)

Foreign investment control

- Foreign investment control has always existed: big capital movements in and out of the country are easy to track
  - Used to be indiscriminate in terms of deal type
- From 80s to early 21st century, big liberalization:
  - Washington consensus led by IMF and European Commission
  - Made it much easier to take over a foreign company
- However, since beginning of 21st century, new generation of foreign investment control coming in
  - Focuses on “national security” interests
  - In France: from defence to… yoghurts

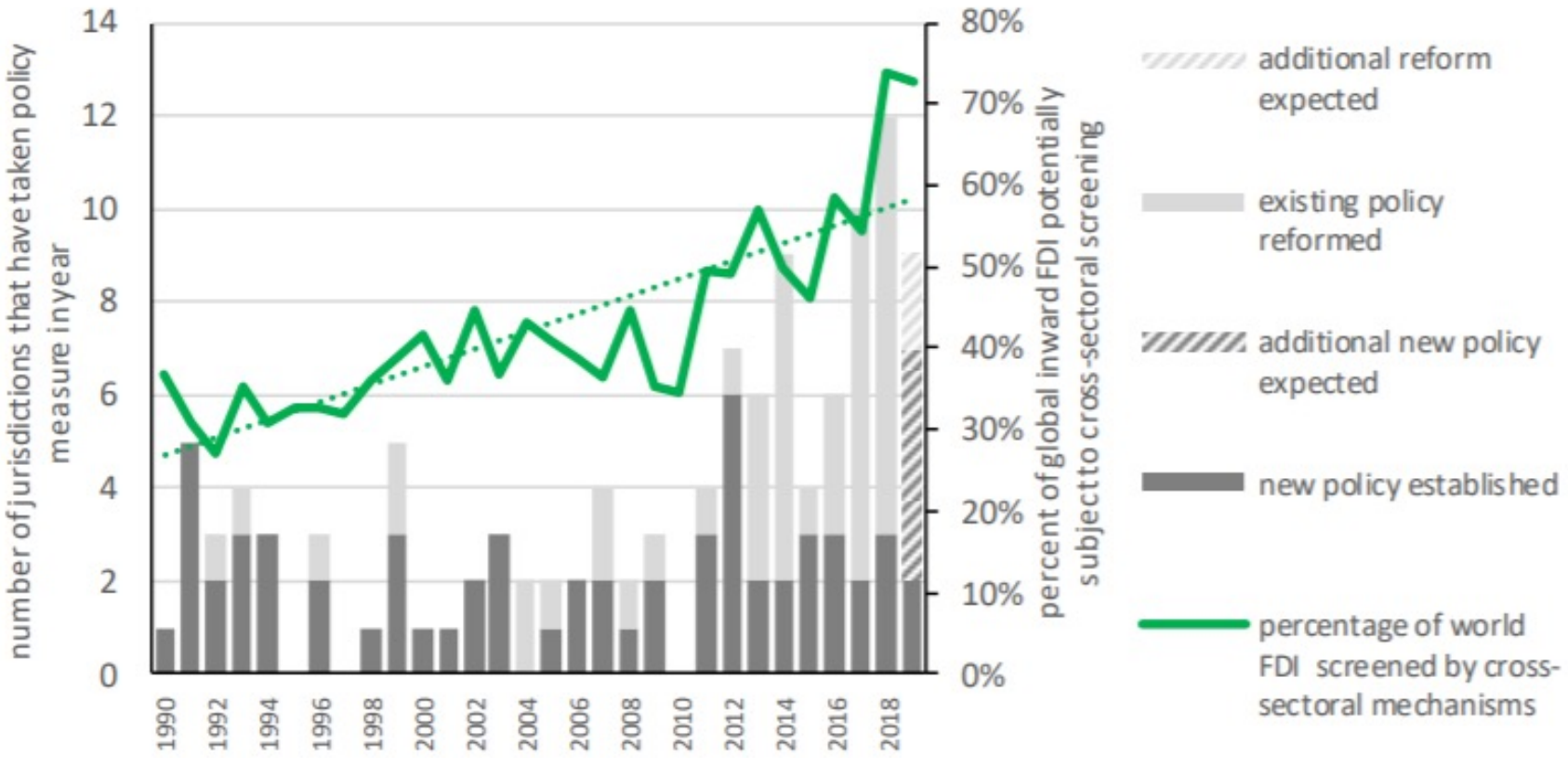
Foreign investment control



Source: OECD calculations based on IMF World Economic Outlook database; IMF Balance of Payments Database; OECD Foreign Direct Investment statistics database.

Foreign investment control

Figure 0.2 Evolution of acquisition-related policies to manage essential security interest since 1990



Note: Based on the currently 62 economies that are invited to OECD-hosted dialogue on international investment policies.
Source: OECD. Some data for 2018 and 2019 are based on estimates and projections.

Foreign investment control

Table 1. Selected FDI screening criteria	
Country	**FDI screening criteria**
Australia	national interest
Austria	public policy and public security, including services of public interest and crisis prevention, which affects a basic interest of society
Belgium (Flanders)[10]	strategic interests of the Flemish Community or the Flemish Region
Canada[11]	net benefit, national security
China	national security, impact on the core key technological innovation development capabilities in important areas of China, national steady economic growth, basic social living order, the R&D capacity of key technologies involving the national security
Germany	public order or security, essential security interests
Finland	key national interest: national defence, public order and security, fundamental interests of society
France	public order, public security or national defense
Hungary	security interests
Iceland	national security, public order, public safety or public health or in the event of serious economic, social or environmental difficulties in particular economic sectors or particular areas
Italy	defense interests, national security, essential interests of the State, order public
Japan	national security, public order, public safety; smooth operation of the economy
Korea, Republic of	national security, national safety and public order, public hygiene or the environmental preservation; Korean morals and customs
Latvia	national security
Lithuania[12]	national security
New Zealand	business experience and acumen, financial commitment, good character of an investor
Norway	national security

source: UNCTAD

Foreign investment control

FDI screening: foreign takeovers over $50 million blocked or abandoned for national security reasons, 2016-September 2019 (cases for which information is publicly available)[19]

Parties	Description	Host-country	Home-country	Value (USD millions)	Sector
Consortium led by Chinese investors Philips NV	The CFIUS raised concerns about a planned sale by the Dutch electronics group Philips of the majority of its Lumiled (United States) LED lights unit to a consortium headed by Go Scale Capital of China on the basis of an alleged threat to the national security of the country. In January 2016, Phillips announced that it was abandoning the proposed sale.	USA	China	3.300	Electronics
Xiamen Sanan Integrated Circuit Co Ltd GCS Holdings	Xiamen Sanan Integrated Circuit announced in March 2016 its intention of acquiring the Taiwan Province of China–based power electronics and chip foundry GCS Holdings Inc, including its California-based subsidiary Global Communication Semiconductors (GCS) LLC. The deal was abandoned on 1 August 2016 because of concerns expressed by the CFIUS.	Taiwan, Province of China / USA	China	226	Technology
Fujian Grand Chip Invest Fund Aixtron SE	The German Ministry of Economy and Energy withdrew its initial certificate of non-objection to the takeover of Aixtron(Germany) by a Chinese company on 24 October 2016. On 2 December 2016, following a recommendation of the Committee on Foreign Investment in the United States (CFIUS), the President of the United States prohibited the acquisition of the United States subsidiary of Aixtron by the same Chinese company on the basis of national security concerns.	Germany / USA	China	723	Technology
Infineon Technologies AG Cree Inc, Wolfspeed	On 16 February 2017, Cree Inc (United States) announced that it would terminate its agreement to sell Wolfspeed, which includes its silicon carbide substrate business, to Infineon Technologies AG (Germany). It stated that "Cree and Infineon have been unable to identify alternatives which would address the national security concerns of the Committee on Foreign Investment in the United States (CFIUS), and as a result, the proposed transaction will be terminated."	USA	Germany	850	Technology

source: UNCTAD

Foreign investment control

FDI screening cases concerning national security, selected countries[20]

(Number of cases per year[21])

		Australia[22]	Canada[23]	France[24]	Germany[25]	Italy[26]	New Zealand[27]	United States[28]
2018	Filed/Examined	-	751	182	n/a	46	-	229
	Approved	633	-	-	-	-	36	-
	Rejected/Blocked	0	1	-	-	-	0	1
	Withdrawn	-	1	-	-	-	-	66
2017	Filed/ Examined	-	737	137	66	30	-	237
	Approved	676	-	-	-	-	22	-
	Rejected/Blocked	0	3	-	-	-	1	1
	Withdrawn	-	0	-	-	-	-	74
2016	Filed/ Examined	-	674	n/a	42	14	-	172
	Approved	645	-	-	-	-	25	-
	Rejected/Blocked	0	1	-	-	-	0	1
	Withdrawn	-	0	-	-	-	-	27
2015	Filed/ Examined	-	719	n/a	40	18	-	143
	Approved	586	-	-	-	-	22	-
	Rejected/Blocked	0	2	-	-	-	0	0
	Withdrawn	-	0	-	-	-	-	13
2014	Filed/ Examined	-	666	n/a	57	8	-	147
	Approved	585	-	-	-	-	15	-
	Rejected/Blocked	1	1	-	-	-	0	0
	Withdrawn	-	0	-	-	-	-	12

Source: ©UNCTAD.

source: UNCTAD

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Session 4:

Dow’s Bid for Rohm and Haas

Overview

1. Rationale
2. Valuation
3. Deal Risks
4. Financial Crisis
5. Choices
6. What happened?

Rationale: Why Buy Rohm?

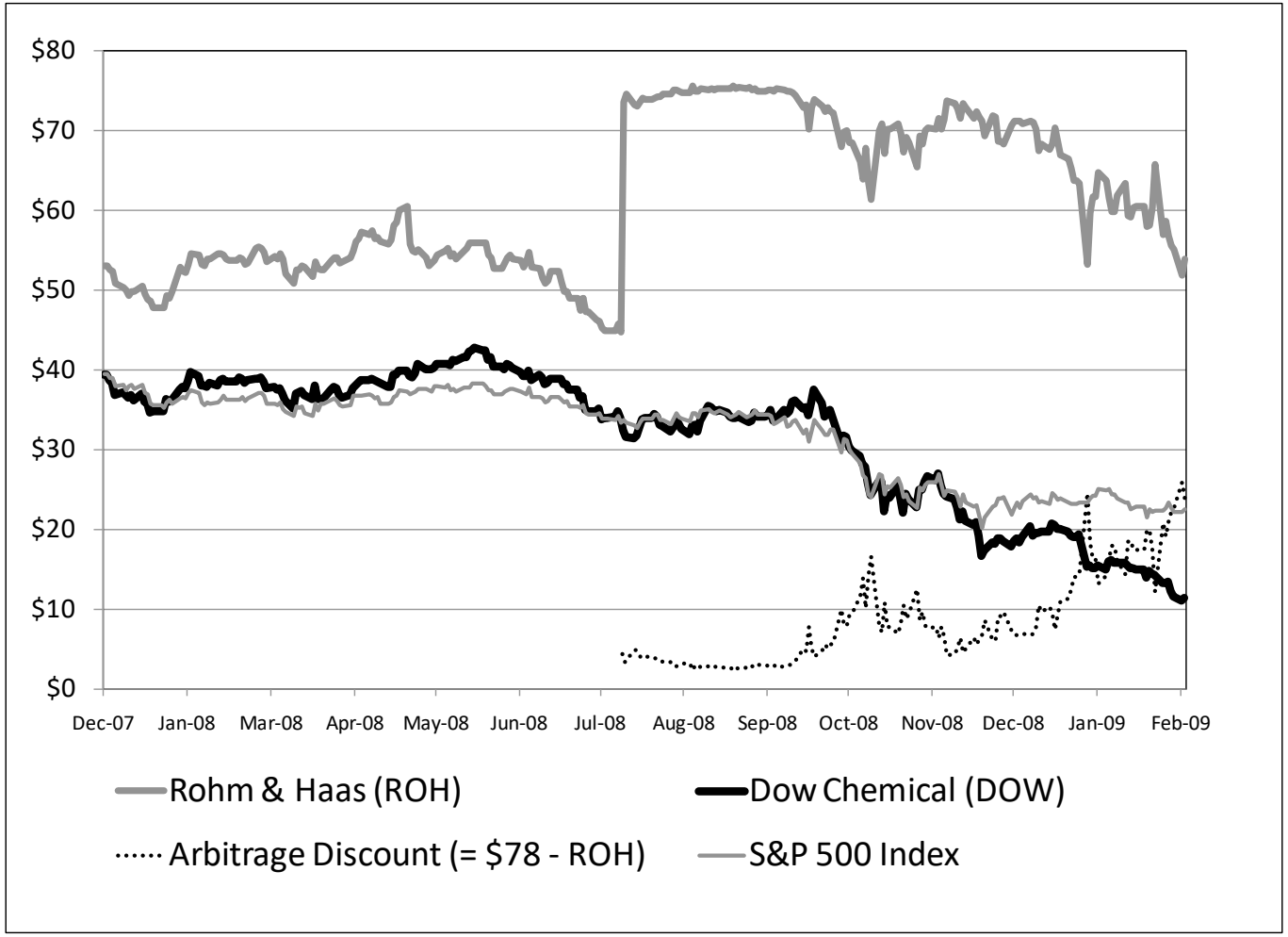
- Cash bid at $78/share in 7/08

- Why is Rohm a "jewel"?
  - Synergy: cost, revenue, financial
  - More growth; less cyclical
  - Rohm is now available

- Negative market reaction
  - 74% Premium==>Overpay?
  - Rohm pre-deal price = $45

Rationale: Why Buy Rohm?

Exhibit 6 Dow Chemical and Rohm and Haas Stock Prices, 12/07 to 2/09



Valuation: Cost of Capital

Weighted Average Cost of Capital (WACC)		Goldman Sachs Assumptions and Sources
Risk-free Rate (R_f)	4.92%	Average 30-year Treasury Yield Over Past 7 Years; Current yield = 4.50%
Equity Beta (β_E)	1.06	Barra's U.S. Equity Model (forward-looking estimate)
Equity Risk Premium (ERP)	5.07%	Ibbotson's Arithmetic Mean (Market - Risk Free Rates) from 1957-2007
Cost of Equity (K_E)	10.3%	Calculated: $K_E = R_f + (\beta_E * ERP)$
Tax Rate	35%	Long-term sustainable tax rate (marginal rate)
Cost of Debt (K_D)	6.1%	Weighted average of rates on Rohm's existing debt as of 12/31/07
Debt / Value Ratio (D/V)	28%	Unstated but implied by the resulting WACC
Equity / Value Ratio (E/V)	72%	Unstated but implied by the resulting WACC
WACC =	**8.5%**	Calculated: WACC = [K_D * (1 - Tax Rate) * (D/V)] + [K_E * (E/V)]
WACC Range	**7.5% - 9.5%**	
Cost of Equity (K_E)	9.9%	Calculated: $K_E = R_f + (\beta_E * ERP)$ and Current Yield
WACC =	**8.2%**	WACC using Current Yield
Asset Beta	**0.76**	Calculated: bA = bE/(1+D/E)
Asset Returns	**8.8%**	Calculated: $K_E = R_f + (\beta A * ERP)$ and Current Yield

Valuation: Rohm Standalone

Valuation of Rohm and Hass on a Standalone Basis (Original FCF Forecast)

WACC =	**8.5%**
Growth Rate =	**2.0%**

	2009	2010	2011	2012	2013
Total Free Cash Flow --Original	**$504**	**$608**	**$747**	**$897**	
Terminal Value				**$14,028**	
TV= (FCF*(1+g))/(WACC-g)					
Present Value Factor	0.921	0.849	0.782	0.721	
Present Value	**$464**	**$516**	**$584**	**$10,761**	

Net Present Value		
NPV of FCF 2009-2012	**$2,212**	**18%**
NPV of Terminal Value	**$10,114**	**82%**
Total NPV	**$12,326**	**100%**
ROH Debt Value	$3,276	
Equity Value	$9,050	
Number of Shares(millions)	195.2	
Equity Value per Share	**$ 46.36**	

Valuation: Sensitivity

Sensitivity Analysis

Discount Rate	Terminal Value Growth Rates					
	0%	1%	2%	3%	4%	5%
6.0%	$56	$69	$88	$120	$185	$377
7.5%	$41	$48	$59	$74	$97	$139
8.2%	$36	$42	$50	$61	$78	$105
8.5%	$34	$39	**$47**	$57	$71	$94
9.5%	$28	$32	$38	$45	$55	$69
11.0%	$21	$24	$28	$33	$39	$47

Valuation: Cost Synergies

Valuation of he Cost Synergies from the Rohm Acquisition

Assumptions:			
Growth Rate	=	2.0%	Assumed
Tax Rate	=	35.0%	Marginal Tax rate
Discount Rate	=	8.80%	Asset Cost of Capital
Annual Cost Synergies	=	$800	Expected Annual Synergies
Restructuring Costs	=	$1 300	Total cost

	2008	2009	2010	2011	2012
Synergies		$400	$600	$800	$816
Restructuring Costs	($650)	($650)			
Sub-total	($650)	($250)	$600	$800	$816
Taxes	($228)	($88)	$210	$280	$286
After-tax CF (b)	($423)	($163)	$390	$520	$530
Terminal Value					$7 800
Discount Factor	1.000	0.919	0.845	0.776	0.714
Present Value	($423)	($149)	$329	$404	$6 056

Net Present Value	**$6 218**
Number of Shares	195.2
Equity Value per Share	**$ 31.85**

Valuation: Growth Synergies

- Growth synergies: $2-2.6 billion in PV
- Nb shares: 195.2 million
- Growth synergies per share = 2300 / 195.2 = $11.8

Valuation: Putting it All Together

	July 2008
Standalone	$47
Synergies:	
Cost	$32
Growth	$12
Financial	$0 (?)
Total	$91

Deal Risks

- Completion
  - Best efforts
  - Ticking fee
  - Break-up fee

Who bears the most important risks?

- Deal/Target Value
  - Consideration (Cash deal)
  - MAE/MAC Clauses

- Enforcement
  - “Specific performance” clause
  - Delaware

What Happened

- What happened? On Feb 12, 2009, Dow cut dividends by 64%.
- This freed up $.9 billion annually to fund debt service
- Negotiated the bridge loan under favorable terms and sold assets, negotiated equity component for acquisition
- Dow stock fell by 81%, from $33.96 before the merger announcement bottoming out at $6.43 on March 9, 2009
- Deal was done at $78.99 per share
- Dow recovered to $36 by February 2011

Bankruptcy Basics

- Bankruptcy is a legal procedure for enforcement or renegotiation of debt contracts that are in default

- In the U.S., a company that has entered bankruptcy gets *automatic stay,* i.e. temporary relief from creditor actions

- Bankruptcy can also be
  - Voluntary = debtor seeks protection (i.e., firm files for protection in a bankruptcy court)
  - Involuntary = creditors file a petition

  (rare, usually the debtor files first)

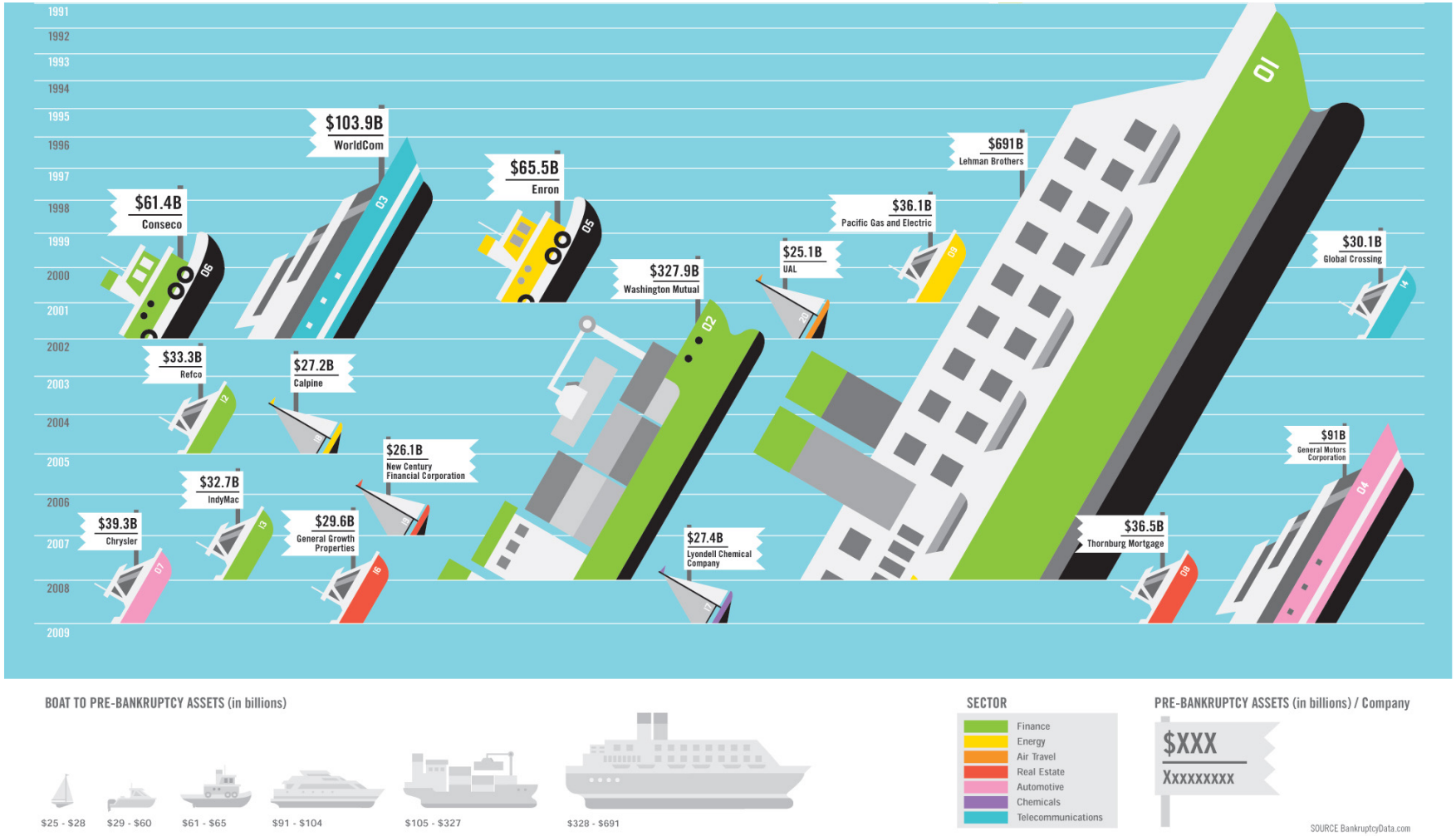
Chapter 7: Liquidation

- Trustee appointed to sell/liquidate assets
- Money distributed according to *strict priority*
  - Secured claims
  - Wages
  - Taxes
  - General unsecured claims (in order of seniority)
  - Equity
- Typically, 4-6 months

Chapter 11: Reorganization

- Debtor gets automatic stay as protection against creditors (to prevent a run on the assets)
- Debtor must present reorganization plan in court (120 days)
- Every class of claimant votes separately
- If plan is accepted, debtor keeps possession of assets, keeps running the firm
  - Court cam “cram down” plan if it deems it fair & equitable

Largest Bankruptcies, 1991-2009



Direct and Indirect CFD

- There are two kinds of costs that a firm can incur as a result of default (together known as costs of financial distress)

- **Direct costs:** Costs incurred as a result of the default
  - I.e., legal and administrative costs, payments to lawyers and accountants, restructuring costs
  - Cost of "fire sales," when assets must be sold immediately
  - Breaking up intangible capital such as human capital or brands

- **Indirect costs:** Costs incurred because of potential future distress
  - Loss of customers due to potential future distress (i.e., as a result of customer concerns about product, warranties, support etc.)
  - Loss of suppliers and employees (or retaining them may be very costly)
  - Fire sales of assets
  - Potential default may introduce agency problems and distort the firm's investments

How Large is Direct CFD?

- Old railroad study by Warner (1977)
  - Direct costs average 5.3% of market value
  - But less than 1% compared to value 7 years earlier and smaller for big railroads
- Broader study of NYSE/AMEX firms 1979-1986 by Weiss (1990)
  - Direct bankruptcy costs = 3.1% of assets (12.4% of debt)
  - Probability of bankruptcy = 0.7% per year
- Direct costs of bankruptcy are small
  - If this cost is a perpetuity, the PV of direct costs can be approximated as V x 4% x 1% / 10% = V x 0.4%
- Compared to the value of the DTS, directs costs are minuscule
  - If bankruptcy matters it must be because of indirect costs of financial distress

Direct and Indirect CFD

What About Indirect CFD?

- **Loss of Customers:** Hesitant to buy product if there is uncertainty about warranties or support
- **Loss of Suppliers:** Hesitant to enter into long-term relationship, ship to inventory, and provide trade credit
- **Loss of Employees:** Concerns about future employment (and pension benefits)
- **Loss of Receivables:** Difficulties collecting from debtors
- **Delayed Liquidation:** May allow inefficient management to remain in control
- **Strategic Weakness:** Competitors may smell blood in the water
- **Agency Conflicts:** Managers may behave differently if firm is close to distress

How Large Are Indirect CFD?

- Indirect costs are difficult to quantify
  - First, identify losses to total (unlevered) firm value
  - Second, identify incremental losses beyond losses arising from economic distress
- One estimate is that indirect costs are 10-20% of firm value: Andrade and Kaplan (1998)

Three Main Conflicts of Interests

1. Wealth transfers from debtholders to shareholders ("cashing out")
2. Risk shifting ("asset substitution")
   - Equity holders' option is worth more when the cash flow is more volatile
3. Debt overhang problem ("underinvestment problem")
   - Can't raise new equity for pos. NPV project if current debt likely to be underwater

- <u>All problems emerge if debt is "risky"</u>